

Experiment 9:

Compare Linear and Polynomial Regression using Python

Python Code:

```
import numpy as np

from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import mean_squared_error, r2_score

X = np.array([[1], [2], [3], [4], [5], [6], [7], [8], [9], [10]])
y = np.array([2, 5, 10, 17, 26, 37, 50, 65, 82, 101])

linear_model = LinearRegression()
linear_model.fit(X, y)

linear_pred = linear_model.predict(X)

poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)
poly_model = LinearRegression()
poly_model.fit(X_poly, y)

poly_pred = poly_model.predict(X_poly)

print("Linear Regression")
print("Predicted values:", np.round(linear_pred, 2))
print("MSE:", round(mean_squared_error(y, linear_pred), 2))
print("R2 Score:", round(r2_score(y, linear_pred), 2))

print("\nPolynomial Regression")
print("Predicted values:", np.round(poly_pred, 2))
print("MSE:", round(mean_squared_error(y, poly_pred), 2))
print("R2 Score:", round(r2_score(y, poly_pred), 2))

new_data = np.array([[7]])

linear_result = linear_model.predict(new_data)
poly_result = poly_model.predict(poly.transform(new_data))

print("\nNew input:", new_data[0][0])
print("Linear Regression prediction:", round(linear_result[0], 2))
```

```
print("Polynomial Regression prediction:", round(poly_result[0], 2))
```

Output:

Linear Regression

Predicted values: [-10. 1. 12. 23. 34. 45. 56. 67. 78. 89.]

MSE: 52.8

R2 Score: 0.95

Polynomial Regression

Predicted values: [2. 5. 10. 17. 26. 37. 50. 65. 82. 101.]

MSE: 0.0

R2 Score: 1.0

New input: 7

Linear Regression prediction: 56.0

Polynomial Regression prediction: 50.0